NUR 774 Chapter 38 Cardiovascular: 135 Hard Questions

70 Select All That Apply + 65 single answer, grouped by topic, from Chapter 38 (heart sounds, murmurs, diagnostics, primary care, specific defects, sudden cardiac death). Each question is followed by its answer, rationale, and a mnemonic.

> Heads-up: Chapter 38 says delay live vaccines 6 months after bypass/blood products. That is different from the 11-month wait after Kawasaki IVIG (2 g/kg). Know both.

How to use: answer the question, then swipe to the next page for the answer, rationale, and mnemonic.

SINGLE ANSWER

An aortic ejection click from aortic stenosis is heard BEST at which location?

A. Apex

B. Left infraclavicular area

C. Lower left sternal border

D. Erb point (left sternal border, 2nd–3rd intercostal space)

Swipe for answer →

Answer 1

D

- **Erb point (left sternal border, 2nd–3rd intercostal space)**

Why: The aortic ejection click is constant in intensity and best heard at Erb point. Erb point also picks up AS and a dilated aortic root.

> 🧠 **"Erb = Ejection click of the Aorta."**

SELECT ALL THAT APPLY

Which conditions are heard at the UPPER LEFT sternal border (ULSB)?

A. Mitral regurgitation

B. Pulmonic valve click of pulmonary stenosis

C. Patent ductus arteriosus

D. Ventricular septal defect

E. Atrial septal defect

F. Pulmonary flow murmur

Swipe for answer →

Answer 2

B, C, E, F

- **Pulmonic valve click of pulmonary stenosis**
- **Patent ductus arteriosus**
- **Atrial septal defect**
- **Pulmonary flow murmur**

Why: ULSB = pulmonic area: PS clicks, pulmonary flow murmurs, ASD, PDA, and venous hum. VSD is at the LLSB; mitral regurgitation is at the apex.

> 🧠 **ULSB = the "P-A-P" corner: Pulmonic stenosis, ASD, PDA.**

SELECT ALL THAT APPLY

Which conditions are heard best at the LOWER LEFT sternal border (LLSB)?

A. Ventricular septal defect

B. Still murmur

C. Tricuspid regurgitation

D. Aortic valve click of aortic stenosis

E. Hypertrophic cardiomyopathy

F. Venous hum

G. Subaortic stenosis

Swipe for answer →

Answer 3

A, B, C, E, G

- **Ventricular septal defect**
- **Still murmur**
- **Tricuspid regurgitation**
- **Hypertrophic cardiomyopathy**
- **Subaortic stenosis**

Why: LLSB: VSD, Still murmur, tricuspid regurgitation, HCM, and subaortic stenosis. Venous hum is in the upper chest; the AS valve click is at the URSB.

> 🧠 **LLSB = "V-STiHS": VSD, Still, Tricuspid regurg, HCM, Subaortic.**

SINGLE ANSWER

A murmur of mitral valve regurgitation is heard best at the:

A. Apex

B. Upper left sternal border

C. Upper right sternal border

D. Erb point

Swipe for answer →

Answer 4

A

- **Apex**

Why: The apex is the mitral area: mitral regurgitation and mitral/aortic clicks.

> 🧠 **"Mitral = the tip (apex) of the heart."**

SELECT ALL THAT APPLY

Which findings belong to the UPPER RIGHT sternal border (URSB)?

A. Aortic valve click of aortic stenosis

B. Still murmur

C. Mitral valve prolapse click

D. Venous hum

E. Ventricular septal defect

Swipe for answer →

Answer 5

A, D

- **Aortic valve click of aortic stenosis**
- **Venous hum**

Why: URSB = aortic area: AS valve clicks and venous hum (venous hum is also heard at the ULSB).

> 🧠 **"Right upper = the Aorta's door."**

SINGLE ANSWER

A 12-year-old's harsh grade III–IV/VI crescendo-decrescendo systolic murmur is loudest at the URSB and radiates to the neck. The most likely lesion is:

A. Ventricular septal defect

B. Aortic stenosis

C. Pulmonic stenosis

D. Still murmur

Swipe for answer →

Answer 6

B

- **Aortic stenosis**

Why: AS murmur: harsh, URSB, radiates to the neck. A thrill may be felt at the suprasternal notch.

> 🧠 **"AS goes up and to the right, then into the neck."**

SINGLE ANSWER

Pulmonic stenosis's systolic ejection murmur classically radiates to:

A. The back and both lung fields

B. The right axilla only

C. The neck only

D. Nowhere; it never radiates

Swipe for answer →

Answer 7

A

- **The back and both lung fields**

Why: PS: harsh mid-to-late SEM at the ULSB, radiating along the LSB, to the neck, back, and both lung fields.

> 🧠 **"PS spreads to the Pulmonary fields (lungs and back)."**

SINGLE ANSWER

Coarctation of the aorta's systolic ejection murmur is best heard in the:

A. Left infraclavicular region, radiating to the back

B. LLSB without radiation

C. Apex, radiating to the axilla

D. URSB, radiating to the neck

Swipe for answer →

Answer 8

A

- **Left infraclavicular region, radiating to the back**

Why: The CoA murmur is heard below the left clavicle, with transmission to the back (where the narrowed aorta is).

> 🧠 **"Coarct = Collarbone (left) to the back."**

SELECT ALL THAT APPLY

Which statements about S1 are correct?

A. It is best assessed at the upper left sternal border

B. It is made by aortic and pulmonic valve closure

C. It is synchronous with the apical and carotid pulses

D. It is the "lubb" of lubb-dupp

E. It is usually loudest at the apex

Swipe for answer →

Answer 9

C, D, E

- **It is synchronous with the apical and carotid pulses**
- **It is the "lubb" of lubb-dupp**
- **It is usually loudest at the apex**

Why: S1 = mitral/tricuspid closure, loudest at the apex, in time with the apical/carotid pulse. S2 is aortic + pulmonic closure and is best assessed at the ULSB.

> 🧠 **"S1 = the Start (lubb), at the tip (apex)."**

SINGLE ANSWER

In a healthy child, how does S2 behave with breathing?

A. It splits with inspiration and becomes single with expiration

B. It splits only with expiration

C. It is widely split and fixed

D. It is always single

Swipe for answer →

Answer 10

A

- **It splits with inspiration and becomes single with expiration**

Why: On inspiration, more blood returns to the right side, so P2 lags behind A2 and S2 splits. On expiration it becomes single.

> 🧠 **"Breathe IN → S2 splits IN two."**

SINGLE ANSWER

Where is S2 best assessed?

A. Apex

B. Upper right sternal border

C. Lower left sternal border

D. Upper left sternal border (pulmonic area)

Swipe for answer →

Answer 11

D

- **Upper left sternal border (pulmonic area)**

Why: S2 splitting is one of the most important parts of the cardiac exam and is best heard in the pulmonic area (ULSB).

> 🧠 **"S2 → the Second rib space, left."**

SELECT ALL THAT APPLY

A child has a LOUD, single S2. Which conditions could cause this?

A. Pulmonary hypertension

B. Pulmonary atresia

C. Innocent Still murmur

D. Transposition of the great arteries

E. Atrial septal defect

Swipe for answer →

Answer 12

A, B, D

- **Pulmonary hypertension**
- **Pulmonary atresia**
- **Transposition of the great arteries**

Why: Pulmonary HTN closes P2 early and accentuates S2. A missing semilunar valve (pulmonary atresia) gives a single S2. In d-TGA, S2 is loud and single because of where the great arteries sit. ASD gives a WIDE FIXED split.

> 🧠 **"Loud and Lonely S2 = pressure (pulm HTN) or a missing/moved valve."**

SINGLE ANSWER

A wide split S2 that does NOT become single on expiration suggests:

A. A normal finding in toddlers

B. Aortic stenosis

C. Increased pulmonary blood flow, as in an atrial septal defect

D. Pulmonary hypertension

Swipe for answer →

Answer 13

C

- **Increased pulmonary blood flow, as in an atrial septal defect**

Why: Extra right-sided volume delays P2 in every breath, so the split stays wide and fixed.

> 🧠 **"ASD = Always Split, Doesn't change."**

SELECT ALL THAT APPLY

Which statements about S3 are correct?

A. It is easier to hear in the left lateral decubitus position

B. It sounds like "Ken-TUCK-y"

C. It is best heard at the apex with the bell

D. It is always pathologic

E. It may be normal in a quiet infant with a fast heart rate

F. It is caused by rapid ventricular filling

Swipe for answer →

Answer 14

A, B, C, E, F

- **It is easier to hear in the left lateral decubitus position**
- **It sounds like "Ken-TUCK-y"**
- **It is best heard at the apex with the bell**
- **It may be normal in a quiet infant with a fast heart rate**
- **It is caused by rapid ventricular filling**

Why: S3 = early diastole, rapid filling, can be normal in children. S4 is the one that is ALWAYS pathologic.

> 🧠 **"S3 = Kentucky: can be a normal Kid. S4 = Tennessee: Trouble."**

SELECT ALL THAT APPLY

Which statements about S4 are correct?

A. It is low-pitched and heard best at the apex with the bell

B. It is best heard with the diaphragm at the ULSB

C. It sounds like "TEN-nessee"

D. It is a normal finding in quiet infants

E. It is always pathologic

F. It occurs in late diastole just before S1

Swipe for answer →

Answer 15

A, C, E, F

- **It is low-pitched and heard best at the apex with the bell**
- **It sounds like "TEN-nessee"**
- **It is always pathologic**
- **It occurs in late diastole just before S1**

Why: S4 reflects forceful atrial contraction into a stiff/distended ventricle. Always pathologic.

> 🧠 **"S4 = Tennessee = Trouble, every time."**

SINGLE ANSWER

A gallop that sounds like "Kentucky" occurs during which part of the cardiac cycle?

A. Early diastole

B. Early systole, immediately after S1

C. Midsystole

D. Late diastole, just before S1

Swipe for answer →

Answer 16

A

- **Early diastole**

Why: S3 is early diastole (right after S2). S4 is late diastole (right before S1).

> 🧠 **"Ken-TUCK-y: the extra beat TUCKs in right AFTER S2."**

SELECT ALL THAT APPLY

Which statements describe a PULMONIC ejection click?

A. It occurs early in systole, right after S1

B. It is constant in intensity

C. It varies with respiration

D. It disappears with inspiration

E. It is heard best at the apex in midsystole

F. It is high in frequency

Swipe for answer →

Answer 17

A, C, D, F

- **It occurs early in systole, right after S1**
- **It varies with respiration**
- **It disappears with inspiration**
- **It is high in frequency**

Why: Pulmonic click varies with breathing and disappears on inspiration. The aortic click is the constant one.

> 🧠 **"Pulmonic click Plays with breathing. Aortic click is Always the same."**

SINGLE ANSWER

A teen has a MIDSYSTOLIC click at the apex that is loudest when standing or leaning forward. The most likely cause is:

A. Aortic stenosis

B. Patent ductus arteriosus

C. Pulmonic stenosis

D. Mitral valve prolapse

Swipe for answer →

Answer 18

D

- **Mitral valve prolapse**

Why: Nonejection clicks are midsystolic at the apex, best heard leaning forward or standing, and are due to MVP.

> 🧠 **"MVP = Middle click, apex, Very upright (standing)."**

SINGLE ANSWER

An ejection click may be confused with which normal-sounding finding?

A. A split first heart sound

B. An S3 gallop

C. A venous hum

D. A split second heart sound

Swipe for answer →

Answer 19

A

- **A split first heart sound**

Why: Ejection clicks occur right after S1 and may sound like a split S1.

> 🧠 **"Click right after lubb → sounds like lubb-lubb."**

SELECT ALL THAT APPLY

Per Box 38.2, which sounds point to a POSSIBLY PATHOLOGIC murmur?

A. Opening snaps

B. S2 that splits with inspiration and becomes single with expiration

C. S4 gallop

D. Fixed splitting of S2 (not from bundle branch block)

E. Systolic clicks

F. An accentuated S2

Swipe for answer →

Answer 20

A, C, D, E, F

- **Opening snaps**
- **S4 gallop**
- **Fixed splitting of S2 (not from bundle branch block)**
- **Systolic clicks**
- **An accentuated S2**

Why: Extra sounds (clicks, snaps, S4) and abnormal S2 (fixed or loud) raise concern. Normal physiologic splitting is reassuring.

> 🧠 **"Extra noises = extra workup."**

SELECT ALL THAT APPLY

Which factors can make ANY murmur louder?

A. Anxiety

B. Anemia

C. Fever

D. Exercise

E. Dehydration with low cardiac output

F. Lying still and calm

Swipe for answer →

Answer 21

A, B, C, D

- **Anxiety**
- **Anemia**
- **Fever**
- **Exercise**

Why: Anything that increases cardiac output intensifies murmurs.

> 🧠 **"FEAR makes murmurs louder": Fever, Exercise, Anemia, Restlessness/anxiety.**

SINGLE ANSWER

Why might a child with a LARGE septal defect have NO murmur?

A. Murmurs are only heard with small hearts

B. There may be little turbulent flow across a large, nonrestrictive opening

C. Large defects always close by 6 months

D. The stethoscope can't pick up septal defects

Swipe for answer →

Answer 22

B

- **There may be little turbulent flow across a large, nonrestrictive opening**

Why: Murmurs come from turbulence. Large nonrestrictive defects (or a nonrestrictive PDA) may not create turbulence.

> 🧠 **"No noise ≠ no problem."**

SINGLE ANSWER

Up to what percentage of children have an innocent murmur at some point?

A. 20%

B. 5%

C. 100%

D. 80%

Swipe for answer →

Answer 23

D

- **80%**

Why: Up to 80% of children have an innocent (functional) murmur from normal flow through normal structures.

> 🧠 **"8 out of 10 kids hum a little."**

SELECT ALL THAT APPLY

Which characteristics are included when describing a heart murmur?

A. Quality
B. Grade (intensity)
C. Pitch
D. Radiation
E. Blood type
F. Timing in the cardiac cycle
G. Location where loudest

Swipe for answer →

Answer 24

A, B, C, D, F, G

- **Quality**
- **Grade (intensity)**
- **Pitch**
- **Radiation**
- **Timing in the cardiac cycle**
- **Location where loudest**

Why: Table 38.4: timing, location, radiation, quality, duration, pitch, grade.

> 🧠 **"Tell Little Rachel Quietly: Pitch, Duration, Grade" (Timing, Location, Radiation, Quality, Pitch, Duration, Grade).**

SELECT ALL THAT APPLY

Which statements about murmur GRADE are correct?

A. It does not necessarily indicate how severe the problem is

B. Grade VI is the most common innocent murmur

C. It may change from supine to sitting

D. A thrill first appears at grade II

E. A grade II murmur always means a small defect

Swipe for answer →

Answer 25

A, C

- **It does not necessarily indicate how severe the problem is**
- **It may change from supine to sitting**

Why: Intensity ≠ severity; a small VSD can be loud. A thrill begins at grade IV.

> 🧠 **"Loud ≠ bad, soft ≠ safe."**

SINGLE ANSWER

A murmur is loud, heard over a wide area, with a palpable thrill, but CANNOT be heard with the stethoscope barely touching the chest. Grade?

A. VI/VI

B. V/VI

C. IV/VI

D. III/VI

Swipe for answer →

Answer 26

C

- **IV/VI**

Why: IV = loud + thrill. V = heard with the stethoscope barely on the chest. VI = heard without a stethoscope.

> 🧠 **"4 = Feel it (thrill). 5 = barely touching. 6 = no touching."**

SINGLE ANSWER

A murmur can be heard with the stethoscope held just off the chest. What grade is it?

A. V/VI

B. VI/VI

C. III/VI

D. IV/VI

Swipe for answer →

Answer 27

B

- **VI/VI**

Why: Grade VI is heard without the stethoscope touching the chest (rare).

> 🧠 **"6 = Sixth sense, no stethoscope on the chest."**

SELECT ALL THAT APPLY

Which statements about DIASTOLIC murmurs are correct?

A. A decrescendo murmur starting with S2 is often aortic or pulmonic regurgitation

B. Most innocent murmurs are diastolic

C. A mid-diastolic rumble at the apex can come from increased flow with a large VSD or PDA

D. They always indicate cardiac pathology

E. They occur between S1 and S2

Swipe for answer →

Answer 28

A, C, D

- **A decrescendo murmur starting with S2 is often aortic or pulmonic regurgitation**
- **A mid-diastolic rumble at the apex can come from increased flow with a large VSD or PDA**
- **They always indicate cardiac pathology**

Why: Diastolic murmurs occur between S2 and S1 and are always pathologic.

> 🧠 **"Diastolic is always Dangerous."**

SELECT ALL THAT APPLY

Which statements about SYSTOLIC murmurs are correct?

A. Most murmurs heard are systolic

B. Aortic stenosis causes an ejection murmur

C. Systolic murmurs are always pathologic

D. A VSD causes a regurgitant holosystolic murmur

E. They include all innocent murmurs

Swipe for answer →

Answer 29

A, B, D, E

- **Most murmurs heard are systolic**
- **Aortic stenosis causes an ejection murmur**
- **A VSD causes a regurgitant holosystolic murmur**
- **They include all innocent murmurs**

Why: Systolic murmurs can be innocent or pathologic; diastolic murmurs are always pathologic.

> 🧠 **"Systolic = Sometimes fine. Diastolic = Danger."**

SINGLE ANSWER

What is the most common cause of a CONTINUOUS murmur in children?

A. Aortic stenosis

B. Still murmur

C. Patent ductus arteriosus

D. Atrial septal defect

Swipe for answer →

Answer 30

C

- **Patent ductus arteriosus**

Why: A continuous murmur runs from S1 through systole AND diastole. PDA is the most common cause; it must be told apart from a venous hum.

> 🧠 **"Continuous Machinery = PDA."**

SINGLE ANSWER

A continuous murmur below the clavicles disappears when the child lies down or turns the head. This is most likely:

A. Aortic regurgitation

B. Coarctation

C. Venous hum

D. Patent ductus arteriosus

Swipe for answer →

Answer 31

C

- **Venous hum**

Why: A venous hum is innocent and continuous, but disappears with head turning, jugular compression, or lying supine. A PDA murmur cannot be suppressed.

> 🧠 **"Hum goes Home when you lie down."**

SELECT ALL THAT APPLY

Which features are typical of an INNOCENT murmur (Box 38.2)?

A. Normal vital signs and ECG

B. Musical or vibratory quality

C. Grade I–III and localized

D. Associated with a palpable thrill

E. Changes with position

F. Pansystolic timing

G. Short and peaking in the first half of systole

Swipe for answer →

Answer 32

A, B, C, E, G

- **Normal vital signs and ECG**
- **Musical or vibratory quality**
- **Grade I–III and localized**
- **Changes with position**
- **Short and peaking in the first half of systole**

Why: Innocent murmurs are soft, short, systolic, positional, musical, and occur in an otherwise healthy child.

> 🧠 **"7 S's": Soft, Systolic, Short, Single, Small area, Sweet, Sensitive (changes with position).**

SELECT ALL THAT APPLY

Which findings suggest a POSSIBLY PATHOLOGIC murmur?

A. Any diastolic murmur

B. Musical quality that disappears with Valsalva

C. Grade IV/VI or louder

D. Harsh quality

E. A murmur that does not change with position

F. A murmur in a child with trisomy 21

Swipe for answer →

Answer 33

A, C, D, E, F

- **Any diastolic murmur**
- **Grade IV/VI or louder**
- **Harsh quality**
- **A murmur that does not change with position**
- **A murmur in a child with trisomy 21**

Why: Box 38.2 pathologic list includes diastolic, thrill, pansystolic, nonsuppressible continuous, clicks, snaps, fixed/accentuated S2, S4, non-positional, grade ≥ IV, harsh, or a syndrome linked to CHD.

> 🧠 **"Harsh, Loud, Diastolic, Doesn't move, or Down syndrome → refer."**

SELECT ALL THAT APPLY

A family is told their child has an innocent murmur. Which teaching is correct?

A. The child needs dental antibiotic prophylaxis

B. It may be louder with fever, anxiety, pain, or exercise

C. No activity restriction is needed

D. The murmur may come and go

E. The child should avoid competitive sports

Swipe for answer →

Answer 34

B, C, D

- **It may be louder with fever, anxiety, pain, or exercise**
- **No activity restriction is needed**
- **The murmur may come and go**

Why: Reassure: no pathology, no restrictions, no special precautions.

> 🧠 **"Innocent = Normal kid, Normal life."**

SINGLE ANSWER

Which innocent-murmur finding may make the murmur DISAPPEAR?

A. Valsalva maneuver, position change, or gentle jugular pressure

B. Exercise

C. Fever

D. Anemia

Swipe for answer →

Answer 35

A

- **Valsalva maneuver, position change, or gentle jugular pressure**

Why: Box 38.2: innocent murmurs may disappear with Valsalva, position, or gentle jugular pressure. Fever, exercise, and anemia make them louder.

> 🧠 **"Innocent murmurs are shy: change position and they hide."**

SINGLE ANSWER

A 4-year-old has a grade II midsystolic, vibratory, "twangy string" murmur at the LLSB that is louder lying down and softer standing. This is:

A. Venous hum

B. Small VSD

C. Pulmonary flow murmur of infancy

D. Still murmur

Swipe for answer →

Answer 36

D

- **Still murmur**

Why: Still murmur: ages 2–6, LLSB/apex, musical/vibratory, louder supine, decreases standing, does not radiate.

> 🧠 **"Still = Strings (twangy guitar), Supine louder."**

SELECT ALL THAT APPLY

Which conditions are in the differential diagnosis of a Still murmur?

A. Coarctation

B. Patent ductus arteriosus

C. Tetralogy of Fallot

D. Small VSD

E. Idiopathic hypertrophic subaortic stenosis (IHSS)

Swipe for answer →

Answer 37

D, E

- **Small VSD**
- **Idiopathic hypertrophic subaortic stenosis (IHSS)**

Why: Table 38.5 lists small VSD and IHSS as the differential for Still murmur (both are at the LLSB).

> 🧠 **"Still sits at the LLSB next to its look-alikes: VSD and IHSS."**

SINGLE ANSWER

A preterm newborn has a soft murmur from the left upper sternal border radiating to both axillae and the back. It is expected to:

A. Indicate tetralogy of Fallot

B. Require PDA ligation

C. Disappear by about 6 months of age

D. Get louder throughout childhood

Swipe for answer →

Answer 38

C

- **Disappear by about 6 months of age**

Why: Pulmonary flow murmur of infancy (peripheral pulmonary stenosis) is common in newborns, especially preterm, radiates to the axillae and back, and is usually gone by 6 months.

> 🧠 **"PPS = Preemie Pulmonary Swish, gone by 6 months."**

SINGLE ANSWER

A thin-chested 12-year-old has a soft, blowing systolic murmur at the pulmonic area with a normally split S2 and no click or thrill. This is most likely:

A. Pulmonic stenosis

B. Venous hum

C. Pulmonary flow murmur of childhood

D. Atrial septal defect

Swipe for answer →

Answer 39

C

- **Pulmonary flow murmur of childhood**

Why: Pulmonary flow murmur of childhood: thin-chested kids 8–14, soft blowing, normal split S2, no click or thrill. ASD would have a FIXED split; PS would have a click.

> 🧠 **"Thin teen, soft blow, Normal split = Normal flow."**

SELECT ALL THAT APPLY

Which findings describe a VENOUS HUM?

A. Disappears when supine

B. Heard below the clavicles in the upright position

C. Disappears with head turning or jugular compression

D. Heard best at the apex

E. Constant swishing sound

F. Always requires cardiology referral

Swipe for answer →

Answer 40

A, B, C, E

- **Disappears when supine**
- **Heard below the clavicles in the upright position**
- **Disappears with head turning or jugular compression**
- **Constant swishing sound**

Why: Venous hum = innocent continuous murmur; differential is PDA.

> 🧠 **"Hum = Head turn or Horizontal makes it vanish."**

SINGLE ANSWER

A primary care provider plans to refer a child to pediatric cardiology for a murmur. What does the text recommend about ordering routine tests first?

A. Order a CBC, ECG, and MRI first

B. Always order an ECG and chest x-ray before referral

C. Routine tests before referral are not cost-effective; cardiology decides which tests to order

D. Order an echocardiogram first, then refer

Swipe for answer →

Answer 41

C

- **Routine tests before referral are not cost-effective; cardiology decides which tests to order**

Why: The cardiologist can decide more precisely which tests are needed.

> 🧠 **"Referring? Don't pre-test, let the expert pick."**

SINGLE ANSWER

A cyanotic newborn receives 100% oxygen and pinks up, with a rising saturation. This suggests:

A. The test is invalid in newborns

B. The cause is primarily cardiac

C. The cause is primarily pulmonary

D. Tetralogy of Fallot

Swipe for answer →

Answer 42

C

- **The cause is primarily pulmonary**

Why: Hyperoxia test: pinking with 100% O2 = lung disease. Minimal or no improvement = heart disease (the blood is bypassing the lungs).

> 🧠 **"Pinks up = Pulmonary. Stays blue = Heart (Cardiac)."**

SINGLE ANSWER

Why might a CBC be ordered in a child with a murmur?

A. To measure pulmonary vascular resistance

B. To diagnose a VSD

C. To detect a heart rhythm problem

D. To rule out severe anemia or polycythemia as a cause

Swipe for answer →

Answer 43

D

- **To rule out severe anemia or polycythemia as a cause**

Why: Anemia increases flow (louder murmur); polycythemia can reflect chronic cyanosis.

> 🧠 **"CBC checks the blood, not the hole."**

SELECT ALL THAT APPLY

What information does a chest radiograph provide in cardiac evaluation?

A. Cardiac contour

B. Heart size and size of specific chambers

C. Heart rhythm

D. Condition of the lungs

E. Exact location of a septal defect

F. Status of pulmonary blood flow

Swipe for answer →

Answer 44

A, B, D, F

- **Cardiac contour**
- **Heart size and size of specific chambers**
- **Condition of the lungs**
- **Status of pulmonary blood flow**

Why: CXR shows size, shape, pulmonary flow, and lungs. Echo locates defects; ECG shows rhythm.

> 🧠 **"X-ray = Size and Shape."**

SELECT ALL THAT APPLY

Which can be done during cardiac catheterization?

A. Deploy devices to close holes or extra vessels

B. Obtain a heart tissue sample

C. Diagnose CHD at 10 weeks' gestation

D. Screen for hearing loss

E. Open narrowed valves or arteries

F. Measure pulmonary vascular resistance and cardiac output

Swipe for answer →

Answer 45

A, B, E, F

- **Deploy devices to close holes or extra vessels**
- **Obtain a heart tissue sample**
- **Open narrowed valves or arteries**
- **Measure pulmonary vascular resistance and cardiac output**

Why: Cath = diagnostic AND interventional.

> 🧠 **"Cath can Measure, Biopsy, Open, and Plug."**

SINGLE ANSWER

How early can fetal echocardiography diagnose congenital heart disease?

A. At 4 weeks

B. 16–18 weeks' gestation (as early as 10 weeks with high-frequency transvaginal echo)

C. Only after birth

D. At 32 weeks

Swipe for answer →

Answer 46

B

- **16–18 weeks' gestation (as early as 10 weeks with high-frequency transvaginal echo)**

Why: Fetal echo detects CHD, arrhythmias, and hemodynamic changes before birth.

> 🧠 **"Sweet 16 for the fetal heart."**

SINGLE ANSWER

Which ECG pattern is classic for an atrial septal defect?

A. Superior axis between −40 and −160 degrees

B. Right axis deviation with an rSR′ (right bundle branch block) pattern in V1

C. Left ventricular hypertrophy with inverted T waves

D. Delta waves

Swipe for answer →

Answer 47

B

- **Right axis deviation with an rSR′ (right bundle branch block) pattern in V1**

Why: ASD → right-sided volume load: RAD, RAE, RBBB rSR′ in V1. Superior axis = AV septal defect.

> 🧠 **"ASD loads the Right: Right axis, Right bundle."**

SINGLE ANSWER

Which ECG finding is characteristic of an AV septal (endocardial cushion) defect?

A. Prolonged QT

B. Normal axis with LVH only

C. Superior axis (−40 to −160 degrees)

D. Right bundle branch block only

Swipe for answer →

Answer 48

C

- **Superior axis (−40 to −160 degrees)**

Why: AVSD shows a superior QRS axis; RVH is common and the PR is prolonged in 50%.

> 🧠 **"AV canal points UP (superior axis)."**

SELECT ALL THAT APPLY

An infant with CHF is not gaining weight. Which nutrition strategies are appropriate?

A. Formula or fortified breast milk at 24, 27, or 30 kcal/oz

B. Nasogastric tube feeding if the infant fatigues with feeding

C. Diluting formula to reduce fluid volume

D. Limiting feeds to every 6 hours

E. Referral to a nutritionist

Swipe for answer →

Answer 49

A, B, E

- **Formula or fortified breast milk at 24, 27, or 30 kcal/oz**
- **Nasogastric tube feeding if the infant fatigues with feeding**
- **Referral to a nutritionist**

Why: Infants with CHF burn more calories and tire with feeding: increase caloric density and use an NG or G tube if needed.

> 🧠 **"Pack the calories, not the volume: 24-27-30."**

SINGLE ANSWER

An infant had cardiopulmonary bypass surgery and received blood products. When can MMR and varicella be given?

A. Immediately; there is no delay

B. 11 months after surgery

C. Never

D. 6 months after bypass and blood/plasma exposure

Swipe for answer →

Answer 50

D

- **6 months after bypass and blood/plasma exposure**

Why: Ch 38: delay live vaccines 6 months after bypass and RBC/plasma exposure. Other vaccines stay on schedule. (Compare: after 2 g/kg IVIG for Kawasaki, the wait is 11 months.)

> 🧠 **"Bypass → wait 6 for live. IVIG (Kawasaki) → wait 11."**

SELECT ALL THAT APPLY

Which infants qualify for RSV prophylaxis per the chapter?

A. An infant with an innocent Still murmur

B. Infants under 1 year with complicated CHD and CHF

C. A healthy 3-year-old with a repaired small ASD

D. Infants under 1 year with cyanotic CHD

E. Infants under 1 year with CHD and pulmonary hypertension

Swipe for answer →

Answer 51

B, D, E

- **Infants under 1 year with complicated CHD and CHF**
- **Infants under 1 year with cyanotic CHD**
- **Infants under 1 year with CHD and pulmonary hypertension**

Why: AAP: RSV prophylaxis for infants younger than 1 year with cyanotic or complicated CHD, especially with CHF or pulmonary HTN.

> 🧠 **"Under 1, Blue or Burdened (CHF/pulm HTN) → RSV shield."**

SINGLE ANSWER

What is the RSV prophylaxis regimen described in the chapter?

A. 15 mg/kg IV once

B. 15 mg/kg IM every 30 days during RSV season, maximum 5 doses

C. 5 mg/kg IM weekly, maximum 15 doses

D. 50 mg/kg PO daily

Swipe for answer →

Answer 52

B

- **15 mg/kg IM every 30 days during RSV season, maximum 5 doses**

Why: Maximum 5 doses of 15 mg/kg IM monthly during the season.

> 🧠 **"15 every 30, up to 5."**

SINGLE ANSWER

A 45-year-old grandmother will babysit an infant with CHD. Her last Td was 3 years ago. What should she receive?

A. Nothing; Td is still valid

B. Td booster in 7 years

C. MMR only

D. Tdap now, regardless of the interval since her last Td

Swipe for answer →

Answer 53

D

- **Tdap now, regardless of the interval since her last Td**

Why: Anyone older than 19 spending time with infants should get Tdap, regardless of the Td interval, to prevent pertussis.

> 🧠 **"Cocoon the baby: Tdap for every adult around them."**

SELECT ALL THAT APPLY

Which measures help prevent avoidable complications in children with CHD?

A. Good handwashing

B. Restricting all physical activity

C. Withholding all vaccines until after surgery

D. Excellent dental hygiene

E. Seasonal influenza vaccine for the infant and caregivers

F. Avoiding contact with people who have URI symptoms

Swipe for answer →

Answer 54

A, D, E, F

- **Good handwashing**
- **Excellent dental hygiene**
- **Seasonal influenza vaccine for the infant and caregivers**
- **Avoiding contact with people who have URI symptoms**

Why: Preventing respiratory infections and keeping gums healthy (to limit endocarditis) are key.

> 🧠 **"Hands, Flu shot, Floss."**

SELECT ALL THAT APPLY

Which statements about physical activity for children with heart disease are correct?

A. Most children self-limit their activity

B. If the child can talk comfortably during activity, the intensity is likely appropriate

C. Cardiology should be consulted before strenuous sports

D. Children with ventricular arrhythmias may need complete activity restriction

E. All children with CHD must avoid all sports

Swipe for answer →

Answer 55

A, B, C, D

- **Most children self-limit their activity**
- **If the child can talk comfortably during activity, the intensity is likely appropriate**
- **Cardiology should be consulted before strenuous sports**
- **Children with ventricular arrhythmias may need complete activity restriction**

Why: Encourage daily activity. The "talk test" guides intensity. Specific conditions need specific limits.

> 🧠 **"Talk test: if you can talk, you can walk (and play)."**

SINGLE ANSWER

Infant and child CPR certification is recommended for:

A. Only the cardiologist

B. Only parents of children with TOF

C. Only school nurses

D. Anyone caring for a child with a heart condition

Swipe for answer →

Answer 56

D

- **Anyone caring for a child with a heart condition**

Why: The chapter says CPR certification is critical for anyone caring for a child with a heart condition.

> 🧠 **"Heart kid at home? CPR for everyone."**

SELECT ALL THAT APPLY

Which conditions qualify for dental endocarditis prophylaxis (Box 38.3)?

A. Unrepaired cyanotic CHD, including palliative shunts

B. Heart transplant recipient with valve disease

C. Small isolated VSD that was never repaired

D. Prosthetic heart valve

E. CHD repaired with prosthetic material, within 6 months of repair

F. Previous infective endocarditis

Swipe for answer →

Answer 57

A, B, D, E, F

- **Unrepaired cyanotic CHD, including palliative shunts**
- **Heart transplant recipient with valve disease**
- **Prosthetic heart valve**
- **CHD repaired with prosthetic material, within 6 months of repair**
- **Previous infective endocarditis**

Why: Only the highest-risk conditions qualify. A small VSD does not require SBE prophylaxis.

> 🧠 **"Plastic, Past IE, Purple (cyanotic), Patch <6 months, Post-transplant valve."**

SINGLE ANSWER

A child had a residual VSD next to a surgical patch. Prophylaxis is needed:

A. Indefinitely, because there is a residual defect at or adjacent to the prosthetic patch

B. Only before tonsillectomy

C. Never

D. Only for 6 months after surgery

Swipe for answer →

Answer 58

A

- **Indefinitely, because there is a residual defect at or adjacent to the prosthetic patch**

Why: Repaired CHD with residual defects at or near a prosthetic patch or device stays high-risk.

> 🧠 **"Leftover leak by the patch = lifelong cover."**

SINGLE ANSWER

What is the oral dental prophylaxis regimen for a child without a penicillin allergy?

A. Amoxicillin 50 mg/kg daily for 10 days

B. Amoxicillin 90 mg/kg/day divided BID

C. Amoxicillin 50 mg/kg as a single dose 30–60 minutes before the procedure

D. Clindamycin 20 mg/kg after the procedure

Swipe for answer →

Answer 59

C

- **Amoxicillin 50 mg/kg as a single dose 30–60 minutes before the procedure**

Why: Single dose, 30–60 minutes BEFORE the procedure.

> 🧠 **"50 mg/kg, 1 dose, 30–60 minutes before."**

SELECT ALL THAT APPLY

A child who qualifies for prophylaxis had hives (urticaria) with amoxicillin and can take oral medications. Which choices are appropriate?

A. Ampicillin 50 mg/kg IM

B. Azithromycin 15 mg/kg

C. Cephalexin 50 mg/kg

D. Clindamycin 20 mg/kg

E. Amoxicillin 50 mg/kg

Swipe for answer →

Answer 60

B, D

- **Azithromycin 15 mg/kg**
- **Clindamycin 20 mg/kg**

Why: Cephalosporins should NOT be used with a history of anaphylaxis, angioedema, or urticaria to penicillins.

> 🧠 **"Hives to Pen? Skip the Cephs: Clinda or Azithro."**

SELECT ALL THAT APPLY

A child needing dental prophylaxis cannot take oral medication and has no penicillin allergy. Which are acceptable?

A. Ampicillin 50 mg/kg IM or IV

B. Cefazolin or ceftriaxone 50 mg/kg IM or IV

C. Amoxicillin 50 mg/kg PO

D. Azithromycin 15 mg/kg PO

E. Clindamycin 20 mg/kg PO

Swipe for answer →

Answer 61

A, B

- **Ampicillin 50 mg/kg IM or IV**
- **Cefazolin or ceftriaxone 50 mg/kg IM or IV**

Why: Non-oral options without allergy: ampicillin, cefazolin, or ceftriaxone.

> 🧠 **"No mouth? Amp or a Ceph in the muscle or vein."**

SELECT ALL THAT APPLY

After endocarditis prophylaxis is given for a procedure, which findings should raise suspicion for IE?

A. New fever

B. Petechiae

C. Normal heart rate

D. Improved appetite

Swipe for answer →

Answer 62

A, B

- **New fever**
- **Petechiae**

Why: Maintain a high index of suspicion for IE if fever or petechiae appear after any procedure.

> 🧠 **"Fever + spots after a procedure = think IE."**

SELECT ALL THAT APPLY

Which findings should prompt referral to pediatric cardiology?

A. A Still murmur in a thriving 4-year-old

B. A hard-to-classify murmur in a child with poor growth

C. Oxygen saturation less than 95%

D. A pathologic murmur

E. Symptoms of CHF

Swipe for answer →

Answer 63

B, C, D, E

- **A hard-to-classify murmur in a child with poor growth**
- **Oxygen saturation less than 95%**
- **A pathologic murmur**
- **Symptoms of CHF**

Why: These are the chapter's findings suggestive of cardiac disease.

> 🧠 **"Under 95, Failing, Funny murmur, or Failing to grow."**

SINGLE ANSWER

An otherwise well 6-year-old has a new possibly pathologic murmur. How quickly should they be referred?

A. Immediately to the emergency department

B. Only if symptoms develop

C. Timely but not urgent

D. Within 1 hour

Swipe for answer →

Answer 64

C

- **Timely but not urgent**

Why: A child with a murmur who is otherwise doing well is referred in a timely but not urgent manner.

> 🧠 **"Well kid + murmur = soon, not siren."**

SELECT ALL THAT APPLY

Which presentations require referral AS SOON AS POSSIBLE?

A. A child with an arrhythmia and dizziness

B. A teen with chest pain during exertion

C. A newborn with cyanosis and poor feeding

D. A well child whose innocent murmur gets louder with fever

E. A child with syncope

Swipe for answer →

Answer 65

A, B, C, E

- **A child with an arrhythmia and dizziness**
- **A teen with chest pain during exertion**
- **A newborn with cyanosis and poor feeding**
- **A child with syncope**

Why: Exertional chest pain, syncope, arrhythmia, dyspnea, CHF signs, abnormal vital signs, or a symptomatic newborn → ASAP.

> 🧠 **"Pain with play, Passing out, Palpitations, Purple newborn = Prompt."**

SELECT ALL THAT APPLY

Which situations warrant a genetic testing referral for a child with CHD?

A. Child with an innocent murmur and normal development

B. Dysmorphic features

C. Mother with multiple miscarriages

D. Other congenital anomalies

E. Growth retardation

F. Neurocognitive deficits

Swipe for answer →

Answer 66

B, C, D, E, F

- **Dysmorphic features**
- **Mother with multiple miscarriages**
- **Other congenital anomalies**
- **Growth retardation**
- **Neurocognitive deficits**

Why: About 35% of CHD is linked to genetic factors.

> 🧠 **"Heart + something else = Gene check."**

SINGLE ANSWER

During which weeks of gestation does CHD develop?

A. 20th to 24th weeks

B. 2nd to 8th weeks

C. Last trimester

D. 12th to 16th weeks

Swipe for answer →

Answer 67

B

- **2nd to 8th weeks**

Why: This is why exposures early in pregnancy (often before a woman knows she is pregnant) matter.

> 🧠 **"The heart is built in weeks 2–8."**

SELECT ALL THAT APPLY

Which maternal exposures are linked to CHD?

A. Retinoic acid

B. Ibuprofen or naproxen

C. Acetaminophen in normal doses

D. Lithium

E. ACE inhibitors

F. Alcohol

Swipe for answer →

Answer 68

A, B, D, E, F

- **Retinoic acid**
- **Ibuprofen or naproxen**
- **Lithium**
- **ACE inhibitors**
- **Alcohol**

Why: Teratogens listed: lithium, retinoic acid, antiepileptics, NSAIDs, ACE inhibitors, TCAs, sulfonamides, sulfasalazine, tobacco, alcohol, cocaine, marijuana.

> 🧠 **"LIRA ACT: Lithium, Isotretinoin (retinoic acid), NSAIDs (ibuprofen), ACE inhibitors, Alcohol, Cocaine/cannabis, Tobacco."**

SELECT ALL THAT APPLY

Which maternal conditions are associated with CHD?

A. Seasonal allergies

B. Febrile illness such as influenza

C. Diabetes mellitus

D. Rubella

E. Phenylketonuria (PKU)

F. Connective tissue disorders

Swipe for answer →

Answer 69

B, C, D, E, F

- **Febrile illness such as influenza**
- **Diabetes mellitus**
- **Rubella**
- **Phenylketonuria (PKU)**
- **Connective tissue disorders**

Why: Maternal illnesses linked to CHD: diabetes, connective tissue disease, PKU, rubella, febrile illness (especially flu).

> 🧠 **"Mom's DRiP-F: Diabetes, Rubella, PKU, Fever (flu)."**

SINGLE ANSWER

A mother asks whether something she did caused her baby's heart defect. Which response best reflects the chapter?

A. "It was caused by stress during pregnancy."

B. "Most heart defects have no identifiable cause."

C. "It happened during labor."

D. "It was definitely genetic from your side."

Swipe for answer →

Answer 70

B

- **"Most heart defects have no identifiable cause."**

Why: Parents often wrongly blame themselves; most CHD has no identifiable cause. Clear information reduces needless guilt.

> 🧠 **"Not your fault: most have no known cause."**

SELECT ALL THAT APPLY

Which heart defects are associated with trisomy 21 (Down syndrome)?

A. Tetralogy of Fallot

B. VSD

C. PDA

D. AV septal defect

E. ASD

F. Supravalvular aortic stenosis

Swipe for answer →

Answer 71

A, B, C, D, E

- **Tetralogy of Fallot**
- **VSD**
- **PDA**
- **AV septal defect**
- **ASD**

Why: About 50% of children with trisomy 21 have CHD; complete AV canal is classic. Supravalvular AS = Williams.

> 🧠 **"Down = AV canal first, then the D's and TOF."**

SINGLE ANSWER

A girl with Turner syndrome (monosomy X) is most at risk for:

A. Bicuspid aortic valve and coarctation of the aorta

B. AV septal defect

C. Supravalvular aortic stenosis

D. Dextrocardia

Swipe for answer →

Answer 72

A

- **Bicuspid aortic valve and coarctation of the aorta**

Why: Turner: bicuspid AV, CoA (35%), pulmonic stenosis. Also check four-extremity BPs.

> 🧠 **"Turner Turns the aorta: Coarctation and Bicuspid valve."**

SELECT ALL THAT APPLY

Which heart defects are associated with 22q11.2 deletion (DiGeorge syndrome)?

A. Mitral valve prolapse

B. Truncus arteriosus

C. Perimembranous VSD

D. Supravalvular aortic stenosis

E. Interrupted aortic arch

F. Tetralogy of Fallot

Swipe for answer →

Answer 73

B, C, E, F

- **Truncus arteriosus**
- **Perimembranous VSD**
- **Interrupted aortic arch**
- **Tetralogy of Fallot**

Why: DiGeorge = conotruncal (outflow) defects.

> 🧠 **"CATCH-22: Cardiac outflow (TOF, Truncus, Interrupted arch)."**

SINGLE ANSWER

Williams syndrome (7q11.23 deletion) is classically associated with:

A. AV septal defect

B. Transposition of the great arteries

C. Supravalvular aortic stenosis and pulmonic stenosis

D. Coarctation and bicuspid aortic valve

Swipe for answer →

Answer 74

C

- **Supravalvular aortic stenosis and pulmonic stenosis**

Why: Williams = supravalvular AS and pulmonic stenosis.

> 🧠 **"Williams sings ABOVE the valve" (supravalvular).**

SELECT ALL THAT APPLY

Noonan syndrome is associated with which cardiac problems?

A. Tricuspid atresia

B. Hypertrophic cardiomyopathy

C. Truncus arteriosus

D. Aortic root dilation

E. Valvular pulmonic stenosis

Swipe for answer →

Answer 75

B, E

- **Hypertrophic cardiomyopathy**
- **Valvular pulmonic stenosis**

Why: Noonan (PTPN11): valvular PS and HCM. Short stature is common.

> 🧠 **"Noonan = PS + HCM: Pulmonic and thick Heart."**

SELECT ALL THAT APPLY

An infant of a diabetic mother has the highest listed risk of which cardiac problems?

A. VSD

B. Cardiomyopathy

C. Transposition of the great arteries

D. Supravalvular aortic stenosis

E. Coarctation

F. Mitral valve prolapse from Marfan

Swipe for answer →

Answer 76

A, B, C, E

- **VSD**
- **Cardiomyopathy**
- **Transposition of the great arteries**
- **Coarctation**

Why: Infant of diabetic mother: TGA, VSD, CoA (3–5%); cardiomyopathy (10–20%).

> 🧠 **"Sugar Mom → Switched arteries (TGA) and a thick heart."**

SINGLE ANSWER

Which pairing of syndrome and heart defect is correct?

A. Marfan syndrome: tricuspid atresia

B. Williams syndrome: AV canal

C. Trisomy 13: aortic root dilation

D. Holt-Oram syndrome: ASD and VSD

Swipe for answer →

Answer 77

D

- **Holt-Oram syndrome: ASD and VSD**

Why: Holt-Oram (TBX5) = ASD/VSD (with upper-limb anomalies). Marfan = MVP and aortic root dilation. Trisomy 13 = VSD, PDA, dextrocardia.

> 🧠 **"Holt-Oram: Hands + Holes."**

SELECT ALL THAT APPLY

Which are signs of CHF in INFANTS (Box 38.4)?

A. Tachypnea

B. Periorbital edema

C. Orthopnea

D. Poor feeding and poor weight gain

E. Diaphoresis

F. Hepatomegaly

Swipe for answer →

Answer 78

A, B, D, E, F

- **Tachypnea**
- **Periorbital edema**
- **Poor feeding and poor weight gain**
- **Diaphoresis**
- **Hepatomegaly**

Why: Orthopnea and peripheral edema are listed for children and teens, not infants. Infants show periorbital edema.

> 🧠 **"Infant HF = Sweaty, Speedy, Swollen eyes, Stuck weight."**

SELECT ALL THAT APPLY

Which CHF signs are listed for children and teens but NOT infants?

A. Tachycardia

B. Orthopnea

C. Peripheral edema

D. Tachypnea

E. Dyspnea with exertion

Swipe for answer →

Answer 79

B, C, E

- **Orthopnea**
- **Peripheral edema**
- **Dyspnea with exertion**

Why: Older kids show adult-like signs: orthopnea, exertional dyspnea, peripheral edema.

> 🧠 **"Older kids get the grown-up signs."**

SINGLE ANSWER

What is the LARGEST group of children with CHF?

A. Those with excessive left-to-right shunting through unrepaired defects

B. Those with rheumatic fever

C. Those with Kawasaki disease

D. Those with myocarditis

Swipe for answer →

Answer 80

A

- **Those with excessive left-to-right shunting through unrepaired defects**

Why: L→R shunts (VSD, PDA, AV canal) cause most pediatric CHF. The myocardium often copes, but volume overload causes symptoms.

> 🧠 **"Too much to the lungs = most HF in kids."**

SELECT ALL THAT APPLY

Which are traditional pediatric heart failure therapies?

A. Prostaglandin E1 for every child

B. Inotropes

C. Fluid boluses

D. Afterload reducers

E. Diuretics

Swipe for answer →

Answer 81

B, D, E

- **Inotropes**
- **Afterload reducers**
- **Diuretics**

Why: Diuretics, inotropes, and afterload reducers (e.g., ACE inhibitors). β-blockers and aldosterone inhibitors target neurohormones.

> 🧠 **"Dump, Pump, Unload."**

SINGLE ANSWER

Which statement about B-type natriuretic peptide (BNP) monitoring is correct?

A. It diagnoses VSD

B. It is required for every infant with a murmur

C. It may help in biventricular heart failure but is not recommended in single-ventricle disease

D. It is most useful after a Fontan

Swipe for answer →

Answer 82

C

- **It may help in biventricular heart failure but is not recommended in single-ventricle disease**

Why: BNP is secreted by stretched ventricles. It's useful in biventricular HF, not single-ventricle disease.

> 🧠 **"BNP: Both ventricles, yes. One ventricle, no."**

SELECT ALL THAT APPLY

Which lesions typically cause CHF in the FIRST WEEK of life?

A. Critical aortic stenosis

B. Coarctation of the aorta

C. Small VSD

D. Rheumatic fever

E. Hypoplastic left heart syndrome

Swipe for answer →

Answer 83

A, B, E

- **Critical aortic stenosis**
- **Coarctation of the aorta**
- **Hypoplastic left heart syndrome**

Why: Duct-dependent left-sided obstructions present early as the ductus closes. VSD-related CHF appears later (weeks), after PVR falls.

> 🧠 **"Week 1 = Left-side blockages crash as the duct closes."**

SINGLE ANSWER

Which type of ASD is most common?

A. Unroofed coronary sinus

B. Sinus venosus

C. Ostium primum

D. Ostium secundum (mid-septum near the foramen ovale)

Swipe for answer →

Answer 84

D

- **Ostium secundum (mid-septum near the foramen ovale)**

Why: Secundum is most common. Sinus venosus is often linked with anomalous pulmonary venous return; primum is seen in Down syndrome; unroofed coronary sinus is rarest.

> 🧠 **"Secundum = Second door in the Middle (foramen ovale)."**

SINGLE ANSWER

In a child with an ASD, the systolic murmur is caused by:

A. Increased flow across the pulmonary valve

B. Aortic stenosis

C. Blood flowing through the atrial hole

D. Mitral regurgitation

Swipe for answer →

Answer 85

A

- **Increased flow across the pulmonary valve**

Why: The grade I–III crescendo-decrescendo murmur at the pulmonic area comes from extra flow across the pulmonary valve, not from the hole itself.

> 🧠 **"The hole is silent; the valve sings."**

SELECT ALL THAT APPLY

Which findings may be found in a child with an ASD?

A. Thin build or exertional dyspnea

B. Wide, relatively fixed split S2

C. Frequent respiratory infections or pneumonia

D. Murmur first noticed at 2–3 years of age

E. Severe cyanosis at birth

Swipe for answer →

Answer 86

A, B, C, D

- **Thin build or exertional dyspnea**
- **Wide, relatively fixed split S2**
- **Frequent respiratory infections or pneumonia**
- **Murmur first noticed at 2–3 years of age**

Why: Many are asymptomatic; symptoms may increase in late adolescence or adulthood.

> 🧠 **"ASD = A Silent Defect that splits S2."**

SELECT ALL THAT APPLY

Which statements about ASD management are correct?

A. SBE prophylaxis is needed only for 6 months after closure

B. Small defects found in infancy may close on their own

C. Low-dose aspirin is given for 6 months after device closure

D. Exercise restriction is unnecessary

E. Lifelong SBE prophylaxis is required

F. Device closure in the cath lab is possible if the child weighs more than 15 kg

Swipe for answer →

Answer 87

A, B, C, D, F

- **SBE prophylaxis is needed only for 6 months after closure**
- **Small defects found in infancy may close on their own**
- **Low-dose aspirin is given for 6 months after device closure**
- **Exercise restriction is unnecessary**
- **Device closure in the cath lab is possible if the child weighs more than 15 kg**

Why: Large defects are closed after age 1, ideally before school entry.

> 🧠 **"ASD device: 15 kg, 6 months aspirin, 6 months prophylaxis."**

SELECT ALL THAT APPLY

Which are complications of an UNTREATED ASD?

A. Left ventricular outflow obstruction

B. Pulmonary hypertension

C. Right ventricular enlargement and failure

D. Paradoxical emboli

E. Coronary artery aneurysm

Swipe for answer →

Answer 88

B, C, D

- **Pulmonary hypertension**
- **Right ventricular enlargement and failure**
- **Paradoxical emboli**

Why: A clot can cross the hole into the systemic circulation (paradoxical embolus).

> 🧠 **"Right side wears out, clots sneak across."**

SINGLE ANSWER

Which type of VSD is most common?

A. Supracristal

B. Muscular

C. Perimembranous

D. Inlet

Swipe for answer →

Answer 89

C

- **Perimembranous**

Why: Four types: perimembranous (most common), supracristal, inlet, muscular.

> 🧠 **"Peri = Popular."**

SINGLE ANSWER

An infant's VSD murmur becomes HIGHER-PITCHED over several months. This suggests:

A. The defect is getting larger

B. Pulmonary hypertension is developing

C. The defect is getting smaller

D. Endocarditis

Swipe for answer →

Answer 90

C

- **The defect is getting smaller**

Why: Smaller holes create higher-pitched, harsher jets.

> 🧠 **"Higher pitch = Hole shrinking."**

SELECT ALL THAT APPLY

Which findings are expected with a LARGE VSD?

A. Low-pitched holosystolic murmur at the LLSB

B. Signs of CHF after the first weeks of life

C. Diastolic rumble at the apex

D. Thrill along the left sternal border

E. Grade I soft musical murmur that disappears standing

F. S3 or S4 gallop if CHF is present

Swipe for answer →

Answer 91

A, B, C, D, F

- **Low-pitched holosystolic murmur at the LLSB**
- **Signs of CHF after the first weeks of life**
- **Diastolic rumble at the apex**
- **Thrill along the left sternal border**
- **S3 or S4 gallop if CHF is present**

Why: Large VSD = lots of flow (rumble at the apex from increased mitral flow) and CHF as PVR falls.

> 🧠 **"Big VSD: Low pitch, Rumble, Thrill, Failure."**

SELECT ALL THAT APPLY

An infant has a small VSD and no CHF symptoms. Which plan is correct?

A. Monitor every 6 months during the first year

B. SBE prophylaxis is not recommended

C. Immediate surgical closure

D. Expect that many small defects close by about age 4

E. Lifelong activity restriction

Swipe for answer →

Answer 92

A, B, D

- **Monitor every 6 months during the first year**
- **SBE prophylaxis is not recommended**
- **Expect that many small defects close by about age 4**

Why: 30–50% of VSDs are small and most close spontaneously by 4 years.

> 🧠 **"Small VSD: Watch, no pills, often seals by 4."**

SINGLE ANSWER

An infant with a pinhole VSD has a murmur, but the echocardiogram does not show the defect. The nurse understands:

A. Cardiac catheterization is required

B. The murmur must be innocent

C. The echo must be repeated immediately

D. Pinhole VSDs may cause a murmur yet not be visualized on echo

Swipe for answer →

Answer 93

D

- **Pinhole VSDs may cause a murmur yet not be visualized on echo**

Why: The chapter notes that in "pinhole" VSDs a murmur may be present while the defect isn't seen on echo.

> 🧠 **"Pinhole: heard, not seen."**

SINGLE ANSWER

A transcatheter device can be used to close a muscular VSD only if the child weighs more than:

A. 15 kg

B. 25 kg

C. 3 kg

D. 6 kg

Swipe for answer →

Answer 94

D

- **6 kg**

Why: VSD device > 6 kg. Compare: ASD device > 15 kg.

> 🧠 **"VSD 6, ASD 15."**

SELECT ALL THAT APPLY

A COMPLETE AV septal defect includes which components?

A. Primum ASD

B. Inlet VSD

C. Pulmonary stenosis

D. Single (common) AV valve

E. Overriding aorta

Swipe for answer →

Answer 95

A, B, D

- **Primum ASD**
- **Inlet VSD**
- **Single (common) AV valve**

Why: Absence of the endocardial cushion leads to a primum ASD, a common AV valve, and an inlet VSD.

> 🧠 **"Complete canal = Atrial hole + Ventricle hole + One valve."**

SINGLE ANSWER

An infant with trisomy 21 and a complete AV canal shows NO CHF signs but has a loud single S2, a precordial heave, minimal murmur, and desaturation with crying. This suggests:

A. Persistent high pulmonary vascular resistance (pulmonary hypertension)

B. The defect has closed

C. An innocent murmur

D. Mild disease needing no follow-up

Swipe for answer →

Answer 96

A

- **Persistent high pulmonary vascular resistance (pulmonary hypertension)**

Why: Some infants with trisomy 21 keep high neonatal PVR, so they don't overcirculate (no CHF) but develop pulmonary HTN. A quiet murmur here is NOT reassuring.

> 🧠 **"Down + quiet murmur + loud S2 = Pressure, not peace."**

SINGLE ANSWER

When is surgical correction of a COMPLETE AV canal defect usually done?

A. Only if symptoms appear in adolescence

B. Before 6 months of age

C. After age 5

D. At school entry

Swipe for answer →

Answer 97

B

- **Before 6 months of age**

Why: Early repair prevents irreversible pulmonary vascular disease. Partial AV canal is repaired in toddler/preschool years.

> 🧠 **"Complete canal: fix before 6 months."**

SINGLE ANSWER

In a term infant, when does the ductus arteriosus normally close?

A. Only after surgery

B. Functionally in 12–72 hours; permanently in 2–3 weeks

C. Permanently within 1 hour

D. Functionally at 6 months

Swipe for answer →

Answer 98

B

- **Functionally in 12–72 hours; permanently in 2–3 weeks**

Why: Functional closure 12–72 h, permanent closure 2–3 weeks (term).

> 🧠 **"Ductus: 3 days to shut, 3 weeks to seal."**

SINGLE ANSWER

Which infant is at HIGHEST risk for a PDA?

A. A term male infant

B. A premature infant weighing less than 1750 g

C. A post-term infant

D. An infant of an older father

Swipe for answer →

Answer 99

B

- **A premature infant weighing less than 1750 g**

Why: PDA frequency rises with prematurity (45–80% under 1750 g). Female:male ratio is 3:1.

> 🧠 **"Tiny preemie = Patent ductus."**

SINGLE ANSWER

Indomethacin is ordered for a 3-month-old TERM infant with a PDA. The nurse should question it because:

A. It only works in girls

B. Indomethacin is contraindicated and ineffective in term or older infants

C. It must be given orally

D. The dose is too low

Swipe for answer →

Answer 100

B

- **Indomethacin is contraindicated and ineffective in term or older infants**

Why: Indomethacin/ibuprofen closure works only in preterm infants.

> 🧠 **"Indomethacin is for Infants born early only."**

SELECT ALL THAT APPLY

Which statements about PDA findings and management are correct?

A. Families should expect lifelong activity limits

B. Surgical ligation does not require cardiopulmonary bypass

C. Coils or plugs can close many PDAs in the cath lab in children older than 8 months

D. The murmur is continuous immediately after birth

E. After the first weeks, a continuous machinery murmur is heard in the left infraclavicular area

F. SBE prophylaxis is recommended for 6 months after device closure

Swipe for answer →

Answer 101

B, C, E, F

- **Surgical ligation does not require cardiopulmonary bypass**
- **Coils or plugs can close many PDAs in the cath lab in children older than 8 months**
- **After the first weeks, a continuous machinery murmur is heard in the left infraclavicular area**
- **SBE prophylaxis is recommended for 6 months after device closure**

Why: Right after birth, the murmur is soft and systolic. It becomes continuous after the first weeks.

> 🧠 **"PDA: early whisper (systolic), later Machinery."**

SINGLE ANSWER

At approximately what oxygen saturation does cyanosis become VISIBLE?

A. 85%

B. 95%

C. 65%

D. 75%

Swipe for answer →

Answer 102

A

- **85%**

Why: Visible cyanosis appears at about 85%. It is easier to see with polycythemia and harder with anemia or fetal hemoglobin.

> 🧠 **"85 = you start to see blue."**

SELECT ALL THAT APPLY

Cyanosis is LESS readily apparent in which situations?

A. Cold environment only

B. Right-to-left shunting

C. Polycythemia

D. Presence of fetal hemoglobin

E. Anemia

Swipe for answer →

Answer 103

D, E

- **Presence of fetal hemoglobin**
- **Anemia**

Why: Less hemoglobin (anemia) or high-affinity fetal Hb hides cyanosis. Polycythemia makes it more obvious.

> 🧠 **"Anemia and Newborn blood Hide the blue."**

SELECT ALL THAT APPLY

Which statements about Tetralogy of Fallot are correct?

A. It is the most common acyanotic lesion

B. It always presents with severe cyanosis at birth

C. "Pink tets" may have little cyanosis

D. It is more common with 22q11.2 deletion and Down syndrome

E. Cyanosis usually increases over the first months as RVOT obstruction worsens

F. It is the most common cyanotic heart lesion

Swipe for answer →

Answer 104

C, D, E, F

- **"Pink tets" may have little cyanosis**
- **It is more common with 22q11.2 deletion and Down syndrome**
- **Cyanosis usually increases over the first months as RVOT obstruction worsens**
- **It is the most common cyanotic heart lesion**

Why: TOF has a spectrum of severity. Mild cases act like a large VSD, and cyanosis is usually present by 6 months.

> 🧠 **"TOF: Tops the cyanotic list, Turns bluer over time."**

SELECT ALL THAT APPLY

Hypercyanotic ("tet") spells increase at what age and with what triggers?

A. Feeding

B. Lying supine at rest

C. Crying

D. 2–4 months of age

E. Sleeping quietly

F. Defecation

Swipe for answer →

Answer 105

A, C, D, F

- **Feeding**
- **Crying**
- **2–4 months of age**
- **Defecation**

Why: Anything that increases oxygen demand or lowers systemic resistance can trigger a spell.

> 🧠 **"Cry, Eat, Poop → Spell."**

SELECT ALL THAT APPLY

Which interventions are appropriate for a hypercyanotic spell?

A. Subcutaneous morphine as ordered

B. Soothing the infant

C. Knee-chest position

D. Giving a large feeding

E. Oxygen

F. Laying the infant flat and supine

Swipe for answer →

Answer 106

A, B, C, E

- **Subcutaneous morphine as ordered**
- **Soothing the infant**
- **Knee-chest position**
- **Oxygen**

Why: Knee-chest raises systemic resistance, reducing right-to-left shunting and increasing pulmonary blood flow.

> 🧠 **"Tuck, Calm, O2, Morphine (TCOM)."**

SELECT ALL THAT APPLY

Which test findings are expected in TOF?

A. Right ventricular hypertrophy with right axis deviation on ECG

B. Rising hemoglobin and hematocrit over time

C. Rib notching

D. Egg-shaped heart

E. Boot-shaped heart with decreased pulmonary vascular markings

Swipe for answer →

Answer 107

A, B, E

- **Right ventricular hypertrophy with right axis deviation on ECG**
- **Rising hemoglobin and hematocrit over time**
- **Boot-shaped heart with decreased pulmonary vascular markings**

Why: Egg on a string = TGA. Rib notching = coarctation.

> 🧠 **"Boot = TOF, Egg = TGA, Notches = Coarct."**

SELECT ALL THAT APPLY

After TOF repair, which long-term issues need lifelong cardiology follow-up?

A. Pulmonary valve regurgitation

B. Atrial and ventricular arrhythmias

C. Coronary artery aneurysm

D. A widening QRS duration, which raises the risk of VT and sudden death

E. Rheumatic fever

Swipe for answer →

Answer 108

A, B, D

- **Pulmonary valve regurgitation**
- **Atrial and ventricular arrhythmias**
- **A widening QRS duration, which raises the risk of VT and sudden death**

Why: Progressive RV dilation widens the QRS; a QRS ≥ 180 ms raises the risk of VT and sudden death. Cardiology must clear for sports.

> 🧠 **"Fixed TOF still needs: Valve, Rhythm, QRS checks."**

SINGLE ANSWER

A child with polycythemia from cyanotic heart disease is at increased risk for:

A. Bleeding into the joints

B. Anemia

C. Hypoglycemia

D. Cerebral thrombosis (stroke)

Swipe for answer →

Answer 109

D

- **Cerebral thrombosis (stroke)**

Why: Polycythemia boosts oxygen-carrying capacity but thickens the blood. Prevent dehydration.

> 🧠 **"Thick blood → brain clot."**

SELECT ALL THAT APPLY

Which findings are typical of d-transposition of the great arteries?

A. Boot-shaped heart

B. Egg-shaped heart on x-ray

C. Bounding femoral pulses with wide pulse pressure

D. Loud, single S2

E. Cyanosis within the first hour to first day of life

F. Often large for gestational age

Swipe for answer →

Answer 110

B, D, E, F

- **Egg-shaped heart on x-ray**
- **Loud, single S2**
- **Cyanosis within the first hour to first day of life**
- **Often large for gestational age**

Why: d-TGA: aorta from the RV, PA from the LV; parallel circulations. Male:female 3:1.

> 🧠 **"TGA = Turns blue Gigantic baby, Egg heart."**

SELECT ALL THAT APPLY

Which interventions are expected for a newborn with d-TGA?

A. Arterial switch (Jatene) procedure in the first days of life

B. Balloon atrial septostomy

C. Lifelong SBE prophylaxis

D. Indomethacin to close the ductus

E. IV prostaglandin E1

Swipe for answer →

Answer 111

A, B, C, E

- **Arterial switch (Jatene) procedure in the first days of life**
- **Balloon atrial septostomy**
- **Lifelong SBE prophylaxis**
- **IV prostaglandin E1**

Why: Keep mixing (PGE1, septostomy), then switch the arteries. Annual echo follow-up.

> 🧠 **"Open, Mix, Switch."**

SINGLE ANSWER

Without treatment, what is the mortality of d-TGA?

A. 10% by the first year

B. 100% within the first hour

C. Nearly 0% with oxygen alone

D. 50% in the first month and 90% by the first year

Swipe for answer →

Answer 112

D

- **50% in the first month and 90% by the first year**

Why: Untreated d-TGA is usually fatal. With surgery, long-term results are excellent.

> 🧠 **"TGA untreated: half by 1 month, nearly all by 1 year."**

SINGLE ANSWER

A child with hypoplastic left heart syndrome is undergoing staged palliation. Match the procedure to its timing:

A. Bidirectional SVC-to-PA anastomosis at 4–6 months; Fontan (IVC-to-PA) at 2–4 years

B. Both stages within the first week

C. Fontan at birth; Glenn at 10 years

D. Arterial switch at 4–6 months

Swipe for answer →

Answer 113

A

- **Bidirectional SVC-to-PA anastomosis at 4–6 months; Fontan (IVC-to-PA) at 2–4 years**

Why: Stage 1 (newborn shunt) → Glenn (SVC→PA) at 4–6 months → Fontan (IVC→PA) at 2–4 years. Transplant if staging fails.

> 🧠 **"Superior first, Inferior last: S before I, like the alphabet."**

SELECT ALL THAT APPLY

Which are long-term complications of single-ventricle palliation?

A. Protein-losing enteropathy

B. Decreased exercise tolerance

C. Rheumatic heart disease

D. Arrhythmias

E. Thromboembolic events, including stroke

Swipe for answer →

Answer 114

A, B, D, E

- **Protein-losing enteropathy**
- **Decreased exercise tolerance**
- **Arrhythmias**
- **Thromboembolic events, including stroke**

Why: Fontan physiology strains many systems. Families need lifelong support and developmental screening.

> 🧠 **"Fontan Four: Protein loss, Palpitations, Plugs (clots), Poor stamina."**

SELECT ALL THAT APPLY

In tricuspid atresia, which statements are correct?

A. It is acyanotic

B. The right ventricle is small

C. About half of patients also have transposition of the great arteries

D. The tricuspid valve is widely open

E. Blood must cross an ASD to the left atrium

Swipe for answer →

Answer 115

B, C, E

- **The right ventricle is small**
- **About half of patients also have transposition of the great arteries**
- **Blood must cross an ASD to the left atrium**

Why: No RA→RV path. Blood goes across the atrial septum, mixes, and the child is cyanotic.

> 🧠 **"Locked tricuspid → Side door (ASD) → Mixed blue blood."**

SINGLE ANSWER

When is SBE prophylaxis recommended for a child with a single-ventricle defect?

A. Never

B. While the child remains cyanotic

C. Only for 6 months after the Fontan

D. Only before tonsillectomy

Swipe for answer →

Answer 116

B

- **While the child remains cyanotic**

Why: Unrepaired or palliated cyanotic CHD (including shunts) qualifies for prophylaxis.

> 🧠 **"Blue = Bacterial cover."**

SELECT ALL THAT APPLY

Which statements about aortic stenosis are correct?

A. Valvular stenosis is the most common type

B. It mainly causes right ventricular hypertrophy

C. The stenotic valve is usually bicuspid

D. It can cause ischemia and fatal ventricular arrhythmias

E. It causes left ventricular hypertrophy

F. It is about 4 times more common in males

Swipe for answer →

Answer 117

A, C, D, E, F

- **Valvular stenosis is the most common type**
- **The stenotic valve is usually bicuspid**
- **It can cause ischemia and fatal ventricular arrhythmias**
- **It causes left ventricular hypertrophy**
- **It is about 4 times more common in males**

Why: Increased LV pressure load → LVH → oxygen demand outstrips coronary supply → ischemia and arrhythmias.

> 🧠 **"AS: Bicuspid, Boys, Big Left ventricle."**

SELECT ALL THAT APPLY

Which physical findings suggest aortic stenosis?

A. Wide pulse pressure with bounding pulses

B. Thrill at the suprasternal notch

C. Harsh grade III–IV/VI crescendo-decrescendo systolic murmur at the URSB

D. Narrow pulse pressure

E. Faint early systolic click

Swipe for answer →

Answer 118

B, C, D, E

- **Thrill at the suprasternal notch**
- **Harsh grade III–IV/VI crescendo-decrescendo systolic murmur at the URSB**
- **Narrow pulse pressure**
- **Faint early systolic click**

Why: Wide pulse pressure with bounding pulses = PDA (or aortic insufficiency). AS narrows the pulse pressure.

> 🧠 **"AS = Narrow pulse, Neck radiation, Notch thrill."**

SINGLE ANSWER

A child with aortic INSUFFICIENCY (regurgitation) would have which murmur?

A. Holosystolic murmur at the ULSB

B. Continuous machinery murmur

C. Early diastolic blowing murmur from the LLSB to the apex

D. Midsystolic click at the apex

Swipe for answer →

Answer 119

C

- **Early diastolic blowing murmur from the LLSB to the apex**

Why: Aortic regurgitation = early diastolic, decrescendo, blowing. Remember: diastolic = always pathologic.

> 🧠 **"Leaky aortic valve = Diastolic blow."**

SELECT ALL THAT APPLY

Which sports guidance for aortic stenosis is correct?

A. Severe AS: all sports are fine if there are no symptoms

B. Severe AS: avoid competitive or intensive sports

C. Moderate AS: low-intensity sports such as golf or bowling, per cardiology

D. Mild AS: all sports, with annual cardiac exams

E. Mild AS: no sports allowed

Swipe for answer →

Answer 120

B, C, D

- **Severe AS: avoid competitive or intensive sports**
- **Moderate AS: low-intensity sports such as golf or bowling, per cardiology**
- **Mild AS: all sports, with annual cardiac exams**

Why: Severe AS (or moderate with symptoms) risks sudden death from ventricular arrhythmias.

> 🧠 **"Mild = all, Moderate = mellow (golf, bowling), Severe = sit out."**

SINGLE ANSWER

A teen had mechanical aortic valve replacement. Which teaching is essential?

A. No medications are needed

B. Lifelong anticoagulation with warfarin

C. Avoid all vaccines

D. Aspirin for 6 months only

Swipe for answer →

Answer 121

B

- **Lifelong anticoagulation with warfarin**

Why: Mechanical valves are prothrombotic and require warfarin. Tissue valves have limited durability in children.

> 🧠 **"Metal valve = Warfarin forever."**

SELECT ALL THAT APPLY

Which tests detect ventricular arrhythmias in a child with aortic stenosis?

A. 30-day event monitor

B. Hyperoxia test

C. 24-hour Holter monitor

D. Tympanometry

Swipe for answer →

Answer 122

A, C

- **30-day event monitor**
- **24-hour Holter monitor**

Why: Echo is the diagnostic test of choice for AS; Holter or event monitors catch arrhythmias.

> 🧠 **"Holter for the day, Event for the month."**

SELECT ALL THAT APPLY

Which statements about pulmonic stenosis are correct?

A. Critical PS in a newborn can cause cyanosis via right-to-left flow across the foramen ovale

B. The click gets louder with inspiration

C. Short stature is common when PS occurs with Noonan or Turner syndrome

D. Most children are asymptomatic with a murmur found on routine exam

E. A systolic ejection click may decrease with inspiration

Swipe for answer →

Answer 123

A, C, D, E

- **Critical PS in a newborn can cause cyanosis via right-to-left flow across the foramen ovale**
- **Short stature is common when PS occurs with Noonan or Turner syndrome**
- **Most children are asymptomatic with a murmur found on routine exam**
- **A systolic ejection click may decrease with inspiration**

Why: The pulmonic click decreases with inspiration and increases with expiration.

> 🧠 **"Pulmonic click quiets on the inhale."**

SINGLE ANSWER

Balloon valvuloplasty for pulmonic stenosis is indicated when the gradient is greater than:

A. 50 mm Hg

B. 20 mm Hg

C. 100 mm Hg

D. 10 mm Hg

Swipe for answer →

Answer 124

A

- **50 mm Hg**

Why: Balloon valvuloplasty for gradients > 50 mm Hg; surgery if unsuccessful.

> 🧠 **"PS: over 50 → Balloon."**

SINGLE ANSWER

A child with MILD pulmonic stenosis wants to play sports. Which guidance is correct?

A. Treat the child normally without activity limits, but monitor for progression during growth spurts

B. Restrict all physical activity

C. Give lifelong SBE prophylaxis

D. Repair immediately

Swipe for answer →

Answer 125

A

- **Treat the child normally without activity limits, but monitor for progression during growth spurts**

Why: Moderate PS can progress during rapid growth (infancy, adolescence). SBE prophylaxis is not needed unless post-op (6 months) or there is prosthetic material.

> 🧠 **"Mild PS: Play on, but recheck in growth spurts."**

SELECT ALL THAT APPLY

Which findings are consistent with coarctation of the aorta?

A. Leg pain or headaches with exercise

B. Cyanosis of the lips at rest

C. Rib notching on chest x-ray

D. Upper-extremity hypertension

E. Weak or delayed femoral pulses

F. Bounding brachial, radial, and carotid pulses

Swipe for answer →

Answer 126

A, C, D, E, F

- **Leg pain or headaches with exercise**
- **Rib notching on chest x-ray**
- **Upper-extremity hypertension**
- **Weak or delayed femoral pulses**
- **Bounding brachial, radial, and carotid pulses**

Why: Hypertension above the narrowing, hypotension below it. Collateral vessels notch the ribs.

> 🧠 **"Bounding up, Barely down, Bumpy ribs."**

SELECT ALL THAT APPLY

A 6-week-old with severe coarctation would most likely show:

A. Bounding femoral pulses

B. Tachypnea

C. Hypertension in the legs

D. Cool lower extremities

E. Poor feeding

F. Lower-body mottling or pallor

Swipe for answer →

Answer 127

B, D, E, F

- **Tachypnea**
- **Cool lower extremities**
- **Poor feeding**
- **Lower-body mottling or pallor**

Why: Severe infant CoA shows up in the first 6 weeks with heart failure and poor lower-body perfusion.

> 🧠 **"Cold legs + poor feeding baby → check femorals."**

SELECT ALL THAT APPLY

Which statements about coarctation management are correct?

A. Restenosis is more likely if repair occurs before 1 year of age

B. Most children with CoA have a bicuspid aortic valve

C. Blood pressure must be monitored lifelong for recoarctation

D. Lifelong SBE prophylaxis is always required

E. PGE1 maintains or reopens the ductus in critical neonatal coarctation

Swipe for answer →

Answer 128

A, B, C, E

- **Restenosis is more likely if repair occurs before 1 year of age**
- **Most children with CoA have a bicuspid aortic valve**
- **Blood pressure must be monitored lifelong for recoarctation**
- **PGE1 maintains or reopens the ductus in critical neonatal coarctation**

Why: SBE prophylaxis is needed only for 6 months post-op or if prosthetic material is used.

> 🧠 **"Coarct: Fix it, then Follow BP forever."**

SINGLE ANSWER

After coarctation repair, a teen may join competitive sports if:

A. The arm-leg BP gradient is less than 20 mm Hg and peak systolic BP is normal at rest and with exercise

B. At least 1 month has passed

C. The arm-leg gradient is less than 50 mm Hg

D. The child has no headaches

Swipe for answer →

Answer 129

A

- **The arm-leg BP gradient is less than 20 mm Hg and peak systolic BP is normal at rest and with exercise**

Why: Also avoid high-intensity static exercise (weight lifting, wrestling) for the first year after surgery.

> 🧠 **"Under 20 = Good to go; no heavy lifting year 1."**

SINGLE ANSWER

Which ECG change develops in an OLDER child with coarctation?

A. Right ventricular hypertrophy only

B. Left ventricular hypertrophy

C. Superior axis

D. Delta waves

Swipe for answer →

Answer 130

B

- **Left ventricular hypertrophy**

Why: Infants may show RVH; older children develop LVH from hypertension.

> 🧠 **"Older coarct = Left side bulks up."**

SINGLE ANSWER

What is the MOST frequent cause of exercise-related sudden cardiac death in the young?

A. Atrial septal defect

B. Innocent murmur

C. Kawasaki disease

D. Hypertrophic cardiomyopathy

Swipe for answer →

Answer 131

D

- **Hypertrophic cardiomyopathy**

Why: Cardiomyopathies, especially HCM and to a lesser extent arrhythmogenic right ventricular cardiomyopathy (ARVC), are the most common causes.

> 🧠 **"HCM = Heart Collapses Midgame."**

SELECT ALL THAT APPLY

Which ELECTRICAL abnormalities can cause sudden cardiac death?

A. Atrial septal defect

B. Wolff-Parkinson-White syndrome

C. Still murmur

D. Brugada syndrome

E. Congenital long QT syndrome

Swipe for answer →

Answer 132

B, D, E

- **Wolff-Parkinson-White syndrome**
- **Brugada syndrome**
- **Congenital long QT syndrome**

Why: Congenital electrical abnormalities: WPW, long QT, Brugada.

> 🧠 **"Electric WLB: WPW, Long QT, Brugada."**

SINGLE ANSWER

A 13-year-old catcher collapses after a baseball hits his chest. The mechanism is:

A. Rheumatic heart disease

B. Commotio cordis: a blow during the T wave triggers ventricular fibrillation

C. Aortic dissection from Marfan syndrome

D. Hypertrophic cardiomyopathy

Swipe for answer →

Answer 133

B

- **Commotio cordis: a blow during the T wave triggers ventricular fibrillation**

Why: Blunt, nonpenetrating trauma to the midchest at the wrong moment (T wave) → VF. Immediate CPR and AED.

> 🧠 **"Commotio = Chest hit at the T wave."**

SELECT ALL THAT APPLY

Which history findings suggest HCM or ARVC in an athlete?

A. Family history of cardiomyopathy

B. Chest pain or dyspnea with exertion

C. Palpitations

D. A soft murmur that disappears when standing

E. Syncope

Swipe for answer →

Answer 134

A, B, C, E

- **Family history of cardiomyopathy**
- **Chest pain or dyspnea with exertion**
- **Palpitations**
- **Syncope**

Why: HCM may have a dynamic late systolic ejection murmur (louder with standing/Valsalva) and a forceful apical beat. ARVC is more common in males around age 15.

> 🧠 **"Faint, Family, Flutters, Fatigue on effort = screen the heart."**

SELECT ALL THAT APPLY

Which ACQUIRED causes of sudden cardiac death are listed in the chapter?

A. Premature coronary disease from familial hypercholesterolemia

B. Performance-enhancing drugs

C. Myocarditis

D. Wolff-Parkinson-White syndrome

E. Commotio cordis

Swipe for answer →

Answer 135

A, B, C, E

- **Premature coronary disease from familial hypercholesterolemia**
- **Performance-enhancing drugs**
- **Myocarditis**
- **Commotio cordis**

Why: WPW is congenital and electrical; the others are acquired.

> 🧠 **"Acquired = Germs, Drugs, Fat, Blows."**